MLOPS

Machine Learning Operations

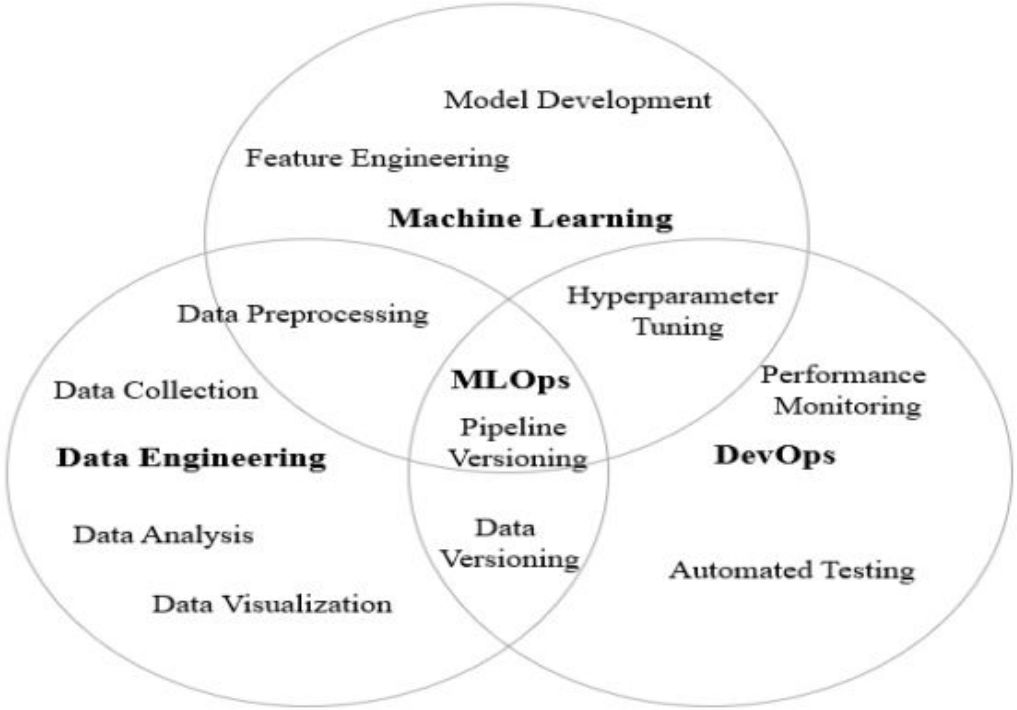
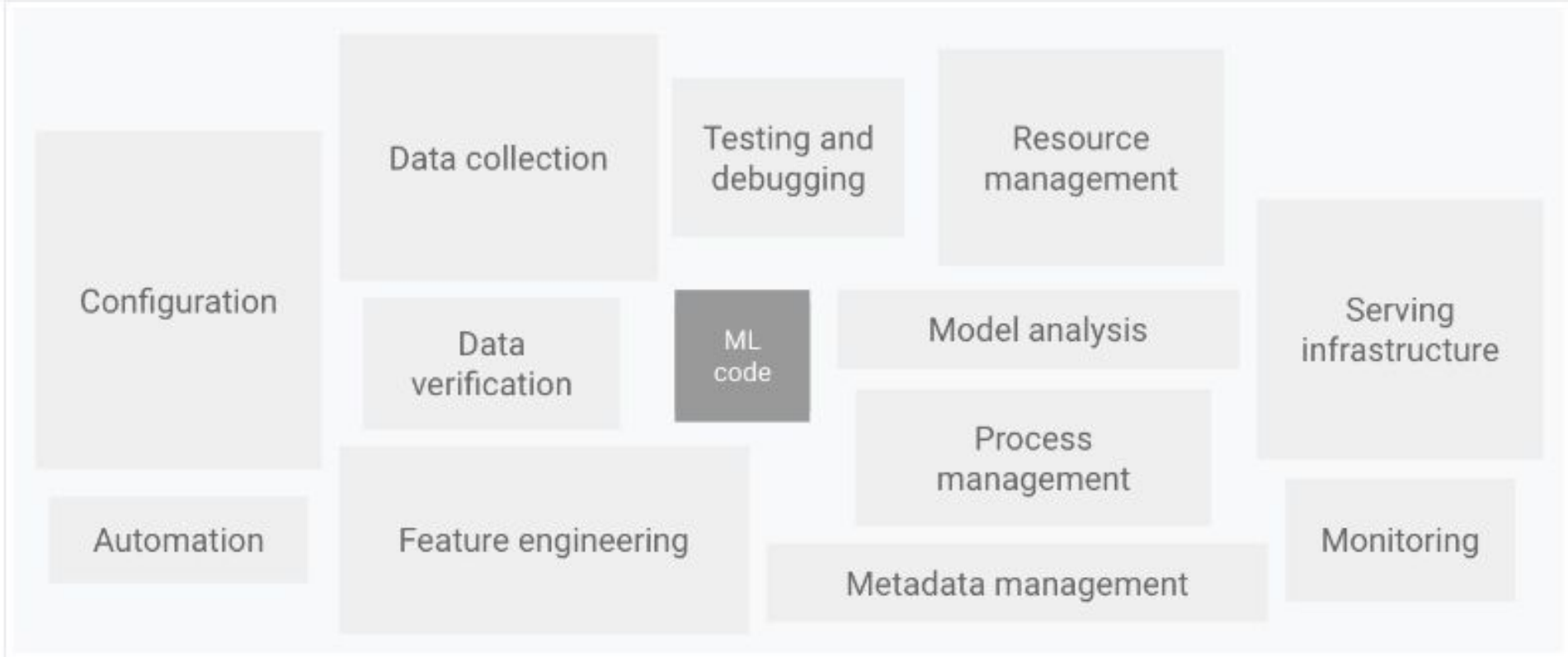


Figure 1.2 MLOps Combination

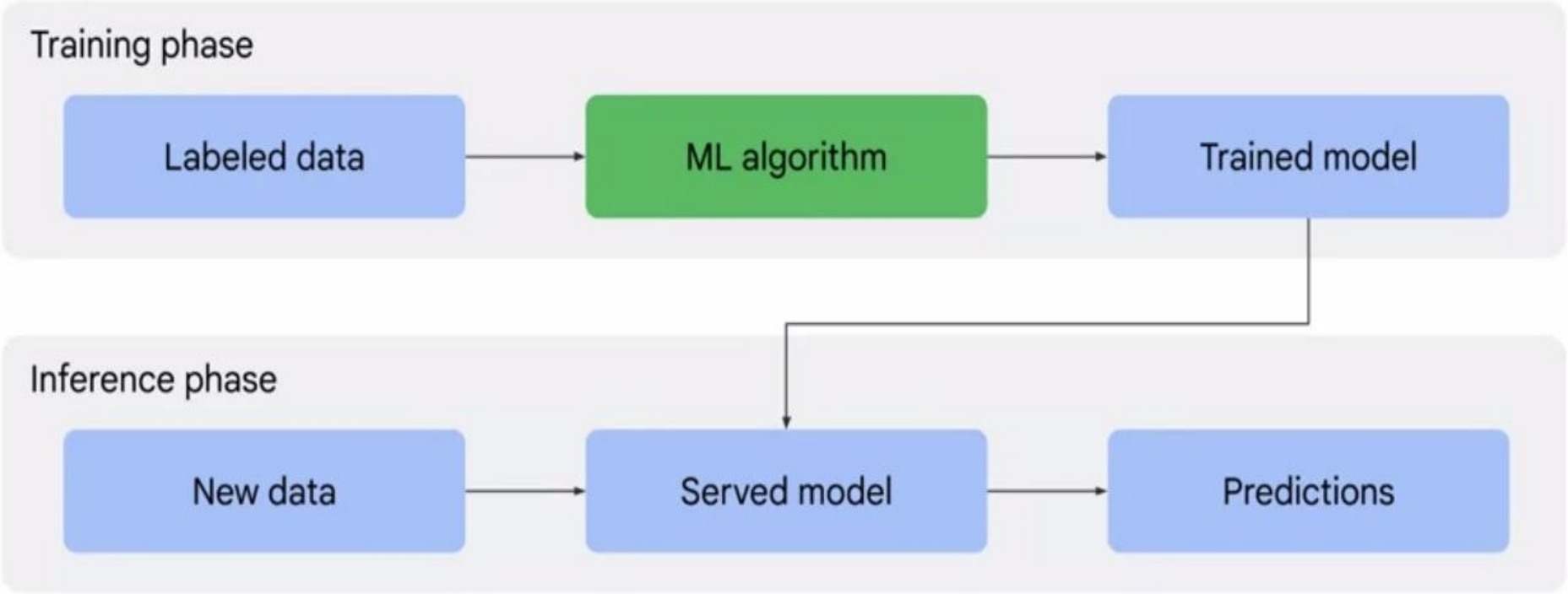
Different roles throughout the ML lifecycle

Product manager	Data analyst	Data engineer	Data scientist	ML developer	ML engineer
Defines insights and objectives.	Queries and analyzes.	Gets clean, useful data.	Verifies that models work.	Creates intelligent apps.	Deploys models into production.

A small fraction of a real-world ML system is composed of the ML code



Operationalizing a model means balancing time, resources, and quality throughout the whole system



Some DevOps concepts translate directly to MLOps



Continuous integration (CI)

1. Checkout the code.
2. Complete the task.
3. Validate against the code base.
4. Perform unit testing.
5. Merge the code.



Continuous delivery or Continuous deployment (CD)

1. Build.
2. Test.
3. Release.



Continuous training (CT)

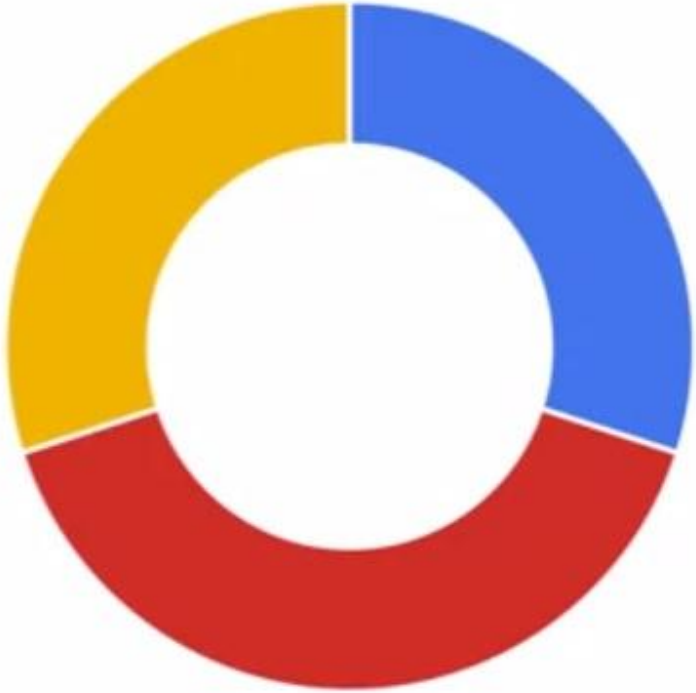
1. Monitor.
2. Measure.
3. Retrain.
4. Serve.

But MLOps differs from DevOps in important ways

	DevOps	MLOps
01	Tests and validates code and components.	Also tests and validates data, data schemas, and models.
02	Focuses on a single software package or service.	Also considers the whole system: the ML training pipeline
03	Deploys code and moves to the next task.	Constantly monitors, retrains, and serves the model.

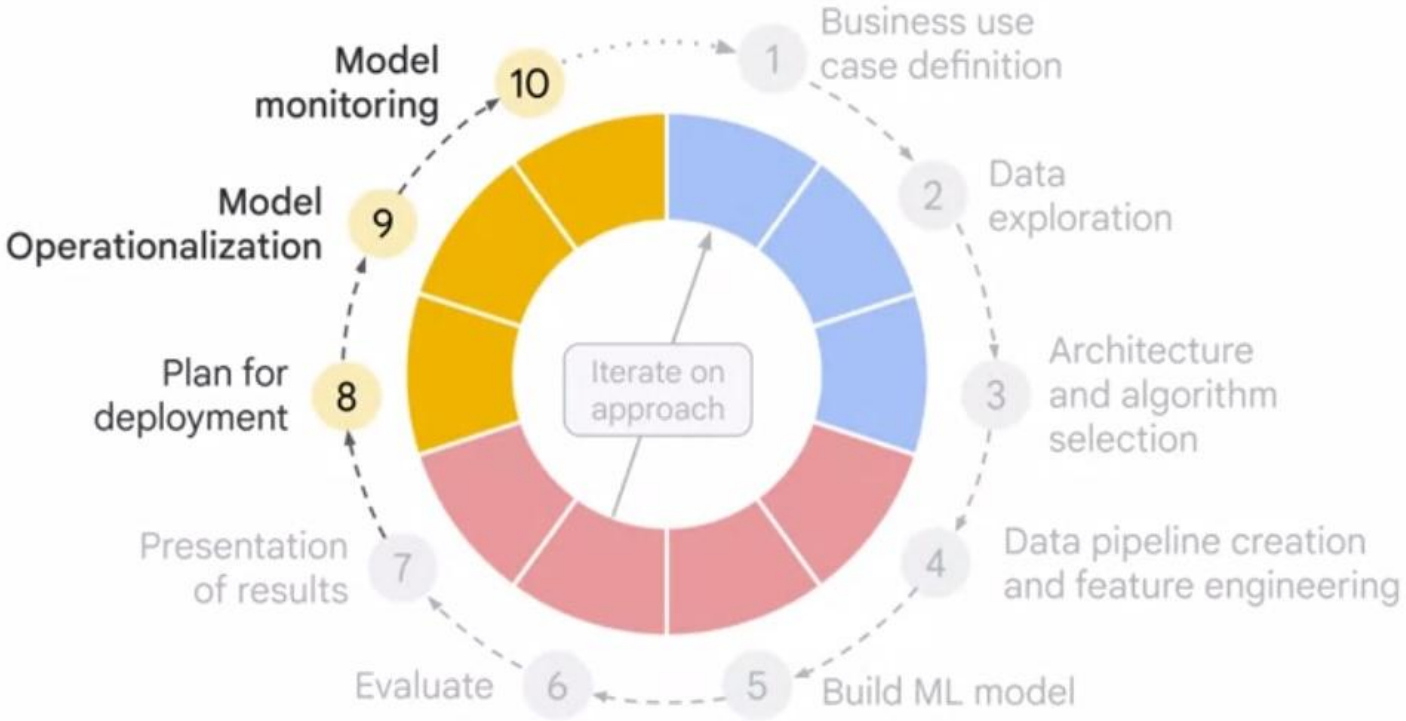
The three phases of the ML lifecycle

1. Discovery phase
2. Development phase
3. Deployment phase



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Model Selection- model fitting-well and it determined what data to be ingested what tool are used which component are required

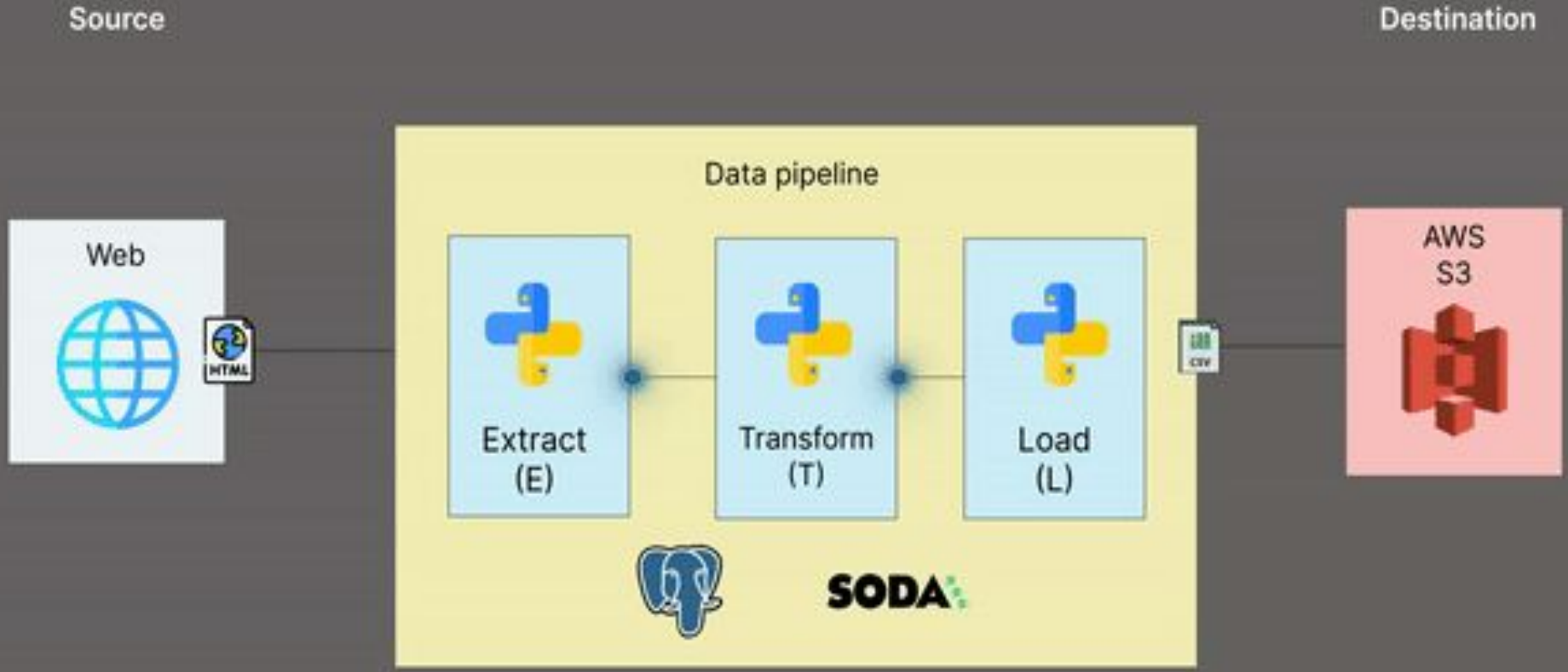
Data Ingestion - capabilities are at the core of any machine learning infrastructure. These capabilities are needed to collect data for model training, application, and refinement.

ML Pipeline Automation numerous tools available that can automate machine learning workflows according to scripts and event triggers. Pipelines are used to process data, train models

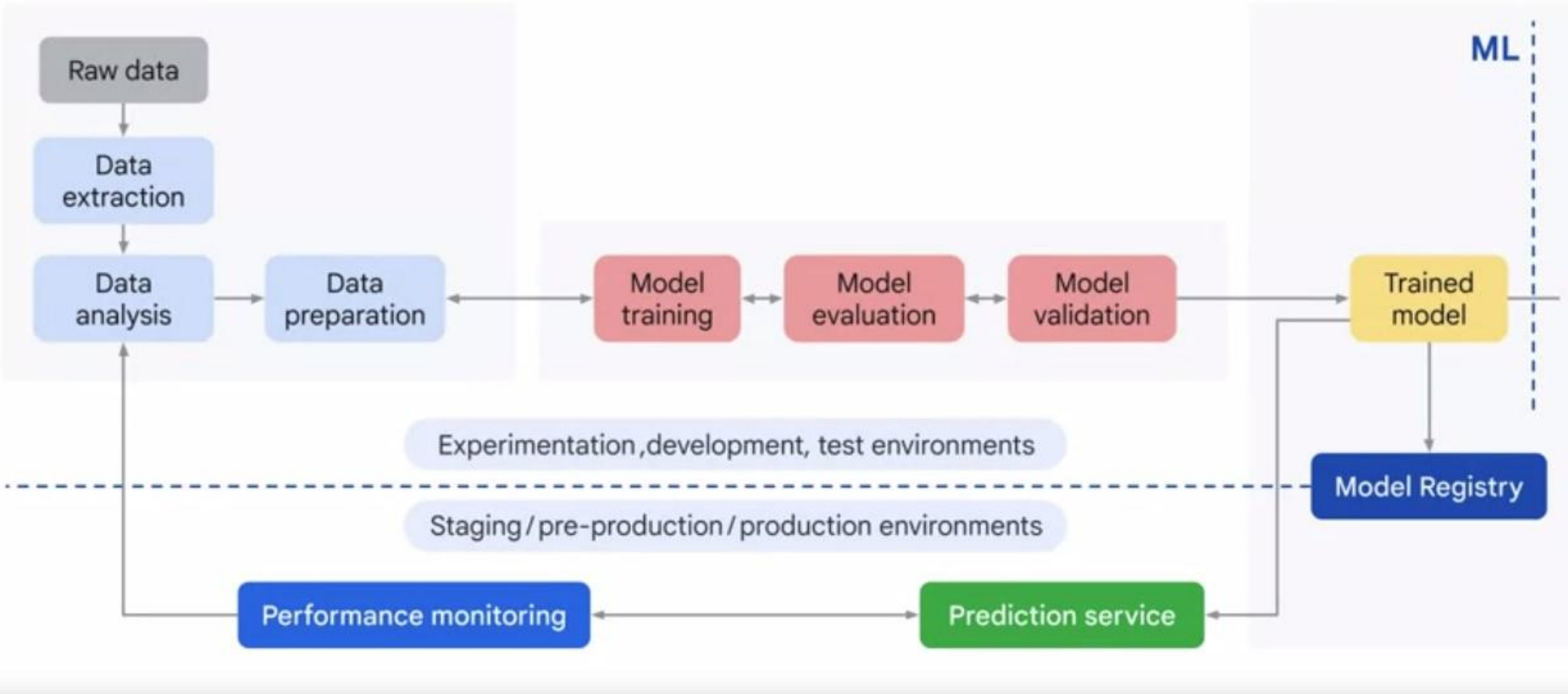
Visualize and Monitoring -are used to gain perspective on how smoothly workflows are moving, how accurate model training is, and to derive insights from model results.

Model Testing - machine learning models requires integrating tooling between training and deployment phases. This tooling is used to run models against manually labeled datasets to ensure that the results are as expected. Thorough testing requires:

Deployment is the final step that you need to account for in your architecture. This step packages your model and makes it available to development teams for integration into services or applications.


Source
Destination
Data pipeline
Web
HTML
Extract
(E)
Transform
(T)
Load
(L)
CSV
AWS
S3
SODA

An ML pipeline contains well-defined processes

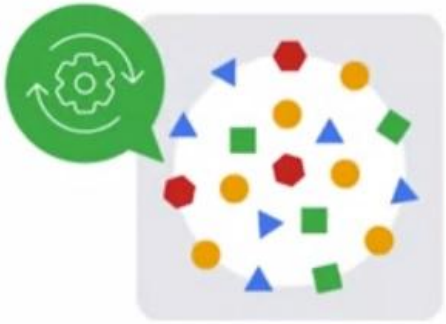


The level of automation defines the maturity of the ML process



Level 0

Build and deploy manually.



Level 1

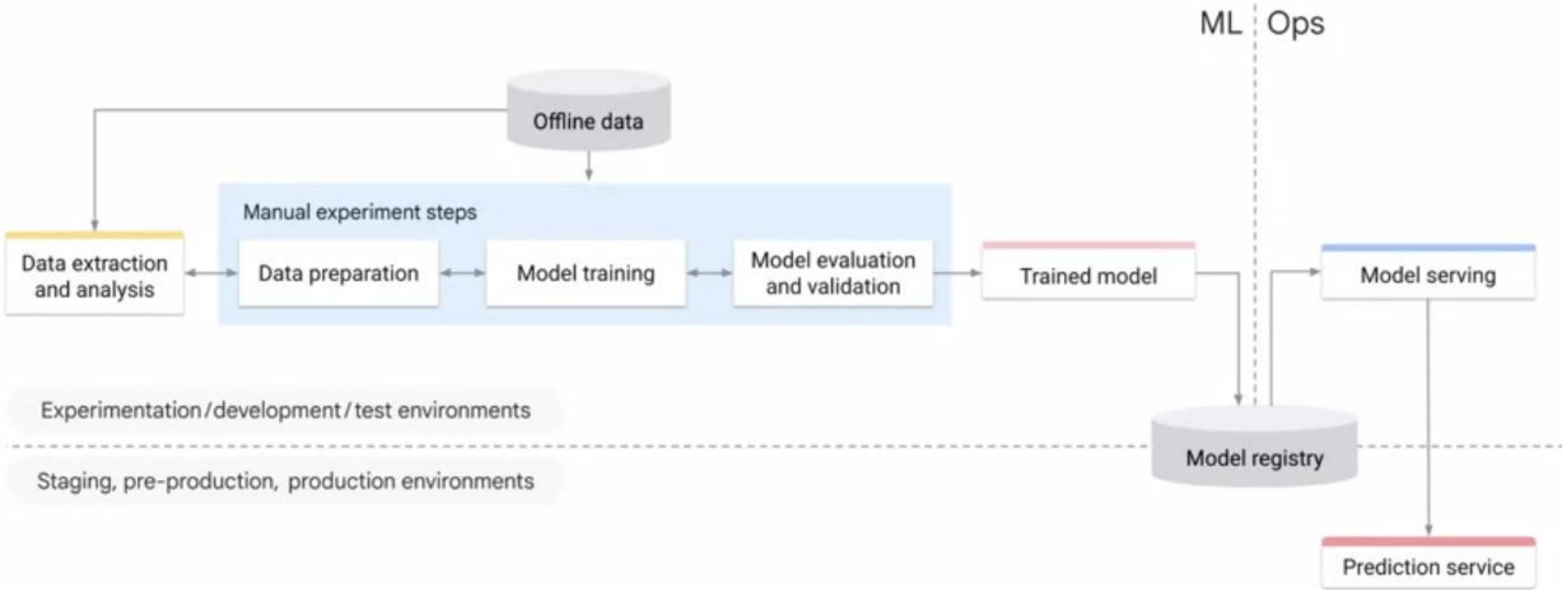
Automate the training phase.



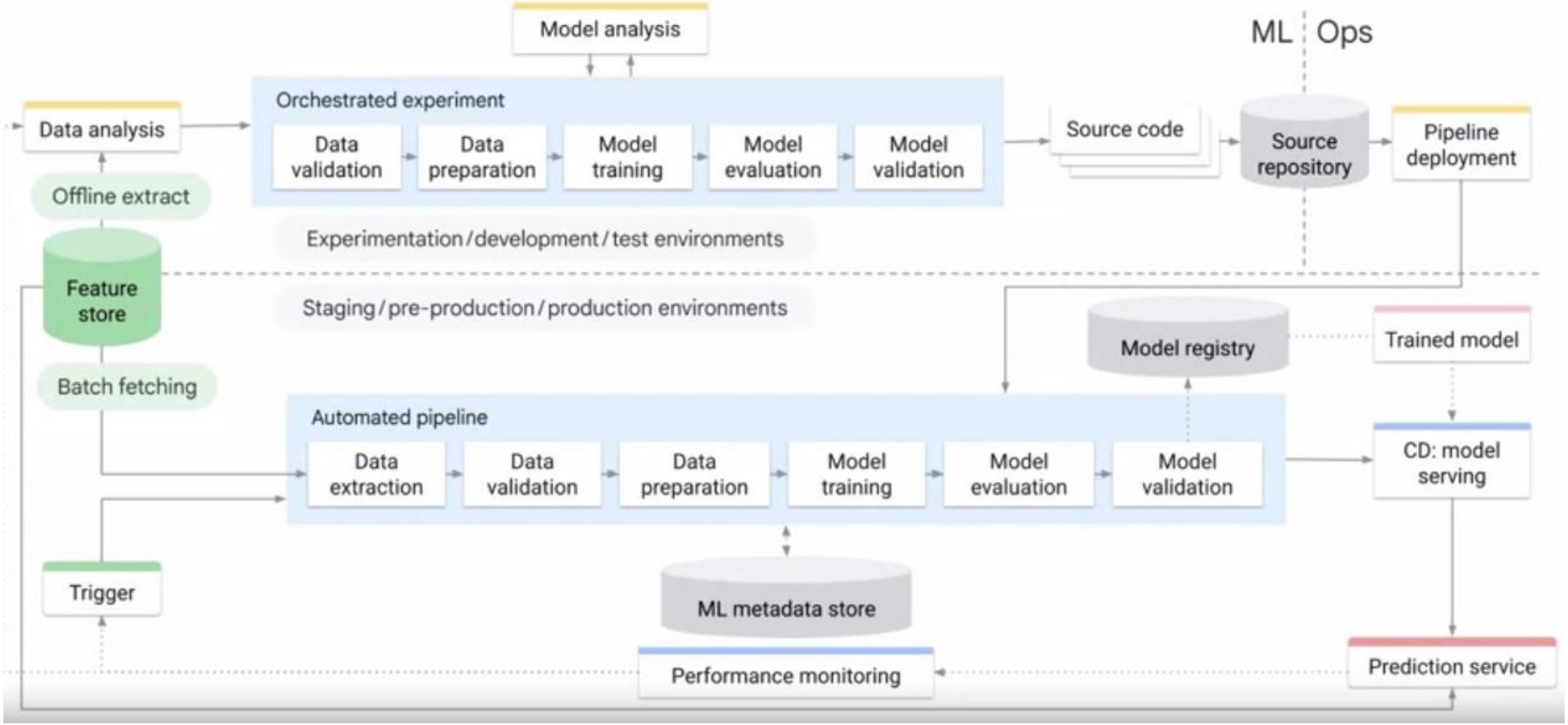
Level 2

Automate training, validation, and deployment.

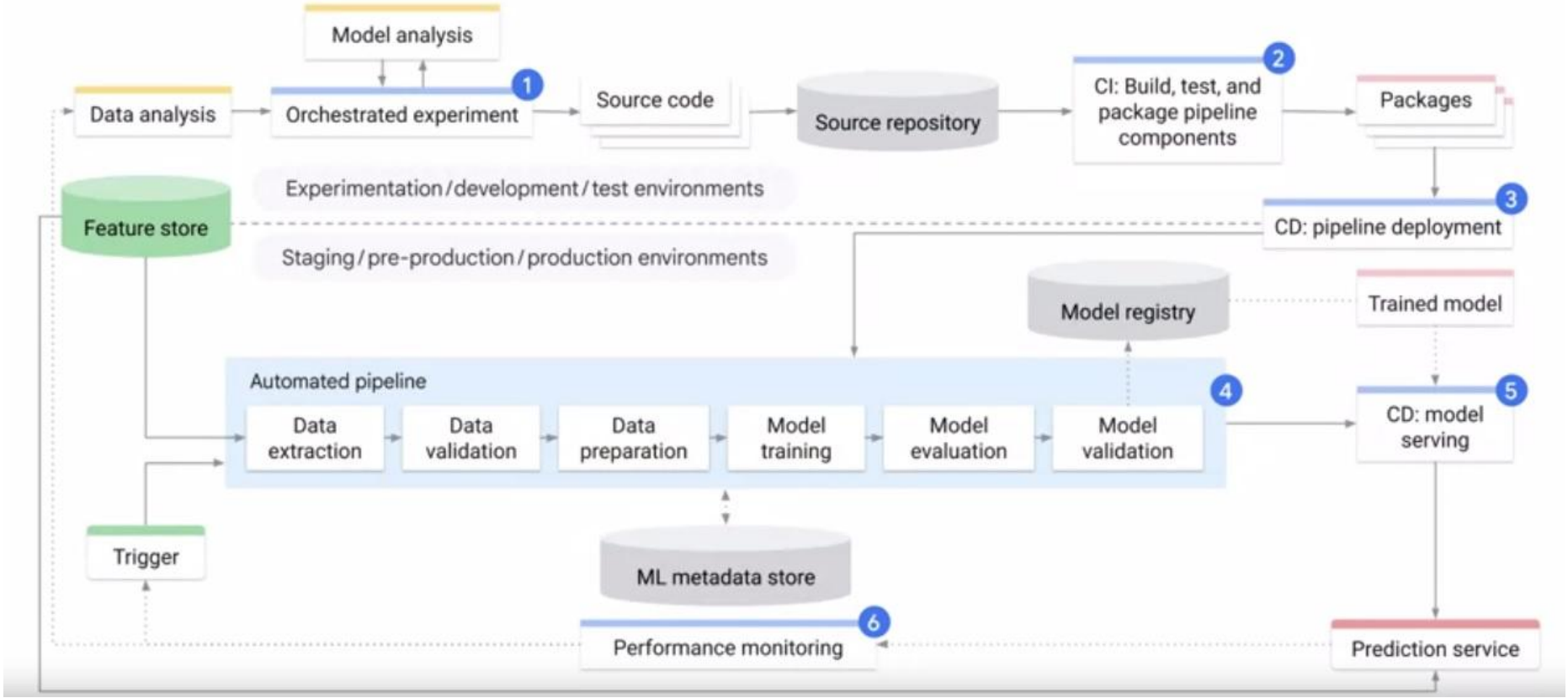
MLOps level 0: Manual process



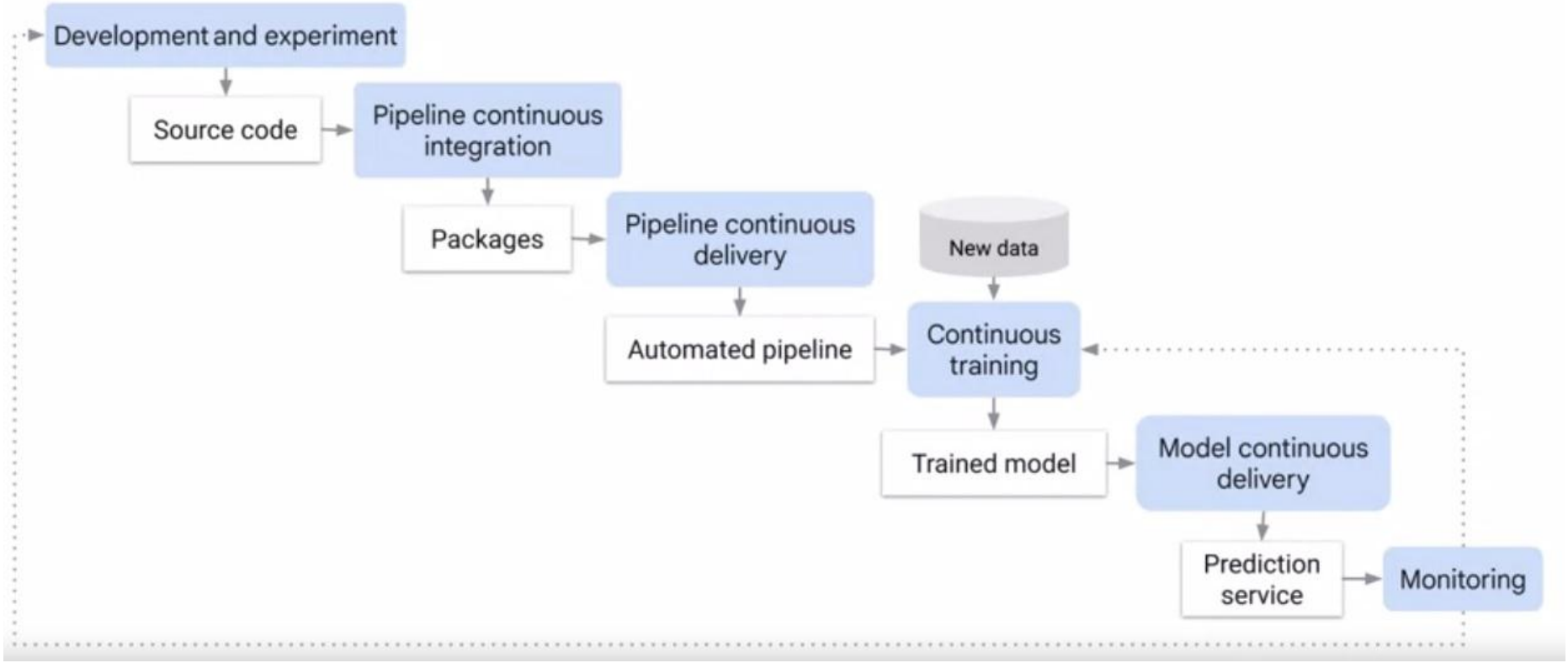
MLOps level 1: ML pipeline automation



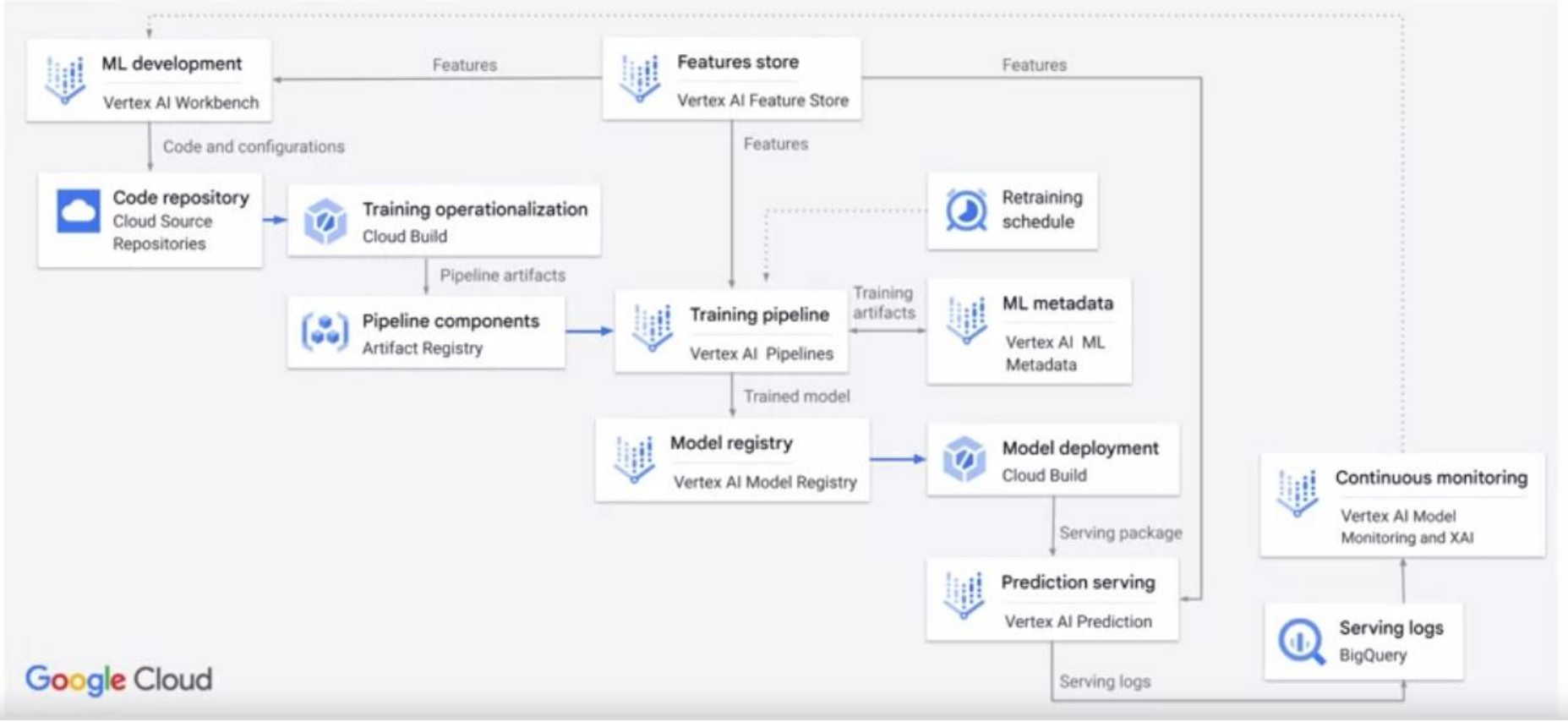
MLOps level 2: CI/CD pipeline automation



MLOps level 2: Characteristics



MLOps on Google Cloud



Applying Devops Practices to Automate ML Systems